

POOJA CHOWDARY MULAGURI

Software Engineer

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SUMMARY

- Software Engineer with 4 years of experience in designing, developing, and deploying scalable applications across finance, e-commerce, and AI-driven solutions.
- Proficient in Python (Advanced), Django, Flask, FastAPI, and React.js, specializing in backend engineering, API development, and cloud computing.
- Experienced in real-time data streaming using Kafka and RabbitMQ, enhancing event-driven architectures, transaction monitoring, and fraud detection.
- Developed and integrated RESTful APIs, GraphQL, OAuth 2.0, and JWT authentication to enable secure and seamless data exchange across platforms.
- Hands-on expertise with AWS (Lambda, EC2, S3, RDS), Azure, and Google Cloud, building cloud-native, cost-efficient, and highly available solutions.
- Skilled in Big Data processing with Apache Spark (PySpark), ETL pipelines using Apache Airflow, and real-time data analytics for business intelligence.
- Strong background in CI/CD automation (Jenkins, GitHub Actions, Terraform, Docker, Kubernetes) for optimized deployments and DevOps workflows.
- Integrated LLM technologies, TensorFlow, PyTorch, and NLP models for AI-driven automation, predictive analytics, and chatbot development.
- Experienced in Agile methodologies (Scrum, TDD), collaborating with cross-functional teams using Jira and Confluence to ensure efficient project execution.
- Passionate about high-performance, secure, and innovative applications, driving business growth, user experience optimization, and data-driven solutions.

SKILLS

- **Programming Languages:** Python (Advanced, Python 3, Type Hinting), JavaScript (ES6), java, Go, HTML, CSS
- **Backend Frameworks:** Django, Flask, Fast API, Node.js (Express.js)
- **Frontend Development:** React.js, Vue.js, Tailwind CSS
- **API Development s Integration:** RESTful APIs, GraphQL, OAuth 2.0, JWT
- **Database Management:** PostgreSQL, MySQL, SQLite, MongoDB, Redis
- **ETL s Data Pipelines:** Apache Airflow, Snowflake, Redshift
- **Cloud s DevOps:** AWS (Lambda, EC2, S3, RDS), Azure, Google Cloud
- **CI/CD s Automation:** Jenkins, GitHub Actions, GitLab CI/CD, Terraform, Docker, Kubernetes
- **Big Data s Streaming:** Apache Spark (PySpark), Kafka, RabbitMQ
- **Machine Learning s AI:** Scikit-learn, TensorFlow, PyTorch, NLP (spaCy, Hugging Face), LLM
- **Testing s Debugging:** pytest, unit test, Pylint, Flake8, Black
- **Security s Compliance:** API Security, OAuth, PyCryptodome
- **Web Scraping s Automation:** Selenium, BeautifulSoup, Scrapy
- **Data Visualization:** Power BI
- **Version Control s Agile Development:** Git, GitHub, GitLab, Agile/Scrum

WORK EXPERIENCE

PNC BANK

Aug 2024 – Present

Software Engineer

- Developed and integrated the Customer Data Integration System, centralizing data from CRM systems, transactions, and customer service records, improving analytics and customer insights, and reducing data retrieval time by 30%.
- Engineered RESTful APIs and Fast API to facilitate seamless data transfer across multiple internal and third-party systems, ensuring high performance and scalability.
- Implemented real-time data streaming using Kafka and RabbitMQ, improving synchronization and providing up-to-date customer insights.
- Utilized AWS S3, RDS, Pandas, and NumPy for efficient data processing, transformation, and cloud-based analytics, enhancing system reliability.
- Built interactive Power BI dashboards, enabling stakeholders to visualize customer interactions and optimize data-driven decision-making.
- Followed Scrum methodologies and adhered to TDD principles, collaborating via Jira to document workflows and automate testing for seamless deployments.

Software Engineer Intern

- Contributed to the Omni-Channel Digital Banking Platform, developing secure and scalable backend services using Django, Fast API, and Node.js, ensuring seamless integration across web and mobile applications.
- Engineered RESTful APIs for customer authentication, transaction processing, and account management, optimizing data flow between frontend applications and backend systems.
- Implemented real-time event streaming using Kafka and RabbitMQ, enhancing transaction monitoring and improving fraud detection capabilities.
- Designed and optimized React.js components for an interactive and responsive UI, ensuring a seamless user experience across multiple devices.
- Leveraged AWS Lambda, S3, and RDS to build a cloud-native infrastructure, ensuring high availability and scalability of banking services.
- Followed Agile methodologies, collaborating with cross-functional teams using Jira and integrating CI/CD pipelines with GitHub Actions, Docker, and Kubernetes for automated deployments.

Infosys BPM

Jul 2021 – December 2022

Software Engineer

- Developed and implemented the Automated Invoice Processing System, utilizing OCR technology (PyTesseract, OpenCV) to extract and process invoice data, reducing manual processing time by 50% and improving approval accuracy.
- Designed and integrated RESTful APIs to streamline data flow between the invoice processing system and financial management software.
- Leveraged Python libraries such as PyTesseract and OpenCV for efficient optical character recognition and image processing.
- Implemented data validation and error handling mechanisms to ensure data accuracy and minimize manual intervention.
- Integrated the system with enterprise databases (PostgreSQL, MySQL) for seamless storage and retrieval of invoice-related data.
- Automated routine tasks such as invoice approval routing and payment processing using custom scripts and workflows.
- Collaborated with cross-functional teams to ensure alignment with business requirements and adherence to regulatory standards.
- Utilized Docker for containerization and deployed the solution on AWS EC2 for better scalability and resource management.
- Conducted performance optimization of the system, reducing invoice processing time by enhancing the OCR accuracy and API response time.

Trigent

January 2020 – June 2021

Associate Software Engineer

- Developed and optimized Django and Flask APIs for handling large-scale data transactions, improving backend efficiency and reducing page load time by 30%, ensuring seamless user interactions.
- Implemented a machine learning-based recommendation engine using Python and Scikit-learn, analysing user behavior and browsing history to deliver personalized product recommendations, increasing customer engagement.
- Designed and deployed real-time order processing and inventory management systems, integrating event-driven architecture to automate stock updates and prevent overselling scenarios.
- Integrated Redis caching strategies to enhance website scalability and response time, reducing database queries and improving application speed, ensuring a smoother shopping experience.
- Built and optimized GraphQL APIs for fetching structured data, reducing network latency and improving data efficiency when interacting with the React-based frontend application.
- Developed dynamic and reusable UI components in React, implementing state management using Redux to enhance user interface responsiveness and create an interactive shopping experience.
- Implemented OAuth 2.0 authentication for secure login and API access, enhancing data security by enforcing token-based authentication and role-based access control for different user levels.

PROJECTS**LLM-Powered Document Summarization and Q&A System**

- Developed an AI-driven system using GPT-4, Lang Chain, and Fast API to process large documents and generate concise summaries.
- Integrated Pinecone and FAISS for efficient vector-based search, enabling rapid retrieval of relevant information.
- Deployed the solution on AWS Lambda and EC2 using Docker for scalability and cost efficiency.

Real-Time Chat Application

- Developed a real-time chat application using React for the front-end and Flask with WebSocket for the back-end to enable instant messaging between users.
- Utilized Redis for message caching and session management to improve performance and ensure quick message retrieval.
- Integrated user authentication using JWT (JSON Web Tokens) for secure login and session management.

EDUCATION**Master of Science in Computer Science**

January 2023 – May 2024

California State University, Long Beach

CERTIFICATIONS

- AWS Certified Developer Associate